Title *Foundation-Model-Based Multimodal Deepfake Detection: A Critical Survey of Architectures, Adaptation Strategies, and Empirical Performance*

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Abstract The proliferation of large-scale generative models (e.g., GANs, diffusion models, neural vocoders) has transformed deepfake creation into a commodity across visual, auditory, and semantic domains. Consequently, forensic research has shifted from monomodal, dataset-specific detectors toward *foundation-model-based* multimodal systems that exploit frozen or lightly-adapted vision-language backbones, self-supervised audio encoders, and multimodal large-language models. This survey systematically reviews the state-of-the-art in such detectors, focusing on (i) the full spectrum of manipulation types (face-swap, reenactment, attribute editing, GAN-/diffusion-generated media, synthetic speech); (ii) architectural families (frozen versus fine-tuned backbones, vision transformers, CLIP-style vision-language models, wav2vec-2.0-style audio encoders); and (iii) cross-modal learning mechanisms (early/late fusion, cross-attention, contrastive alignment, parameter-efficient adaptation such as prompt-tuning, adapters, LoRA). We collate all explicitly reported quantitative results from the retrieved literature (15 numeric records drawn from four papers) and use them to illustrate concrete trade-offs among detection accuracy, robustness to compression and adversarial perturbations, computational cost, and explainability. The analysis reveals persistent gaps—most notably the scarcity of systematic cross-modal generalisation studies, limited robustness reporting, and an absence of calibrated confidence estimates. Building on these observations, we propose a multi-label taxonomy that organises existing work along six dimensions (modality, manipulation type, backbone, adaptation strategy, fusion location, cross-modal objective) and we outline concrete research directions required to advance trustworthy multimodal deepfake forensics.

Keywords Deepfake detection; foundation models; multimodal learning; vision-language models; parameter-efficient adaptation; robustness; explainability; multimedia forensics.

1. Introduction

1.1. Context and Motivation

Generative-AI technologies have matured to the point where high-fidelity synthetic media can be produced with minimal expertise. State-of-the-art generators—ranging from StyleGAN-derived face-swap pipelines to diffusion-based image synthesis and neural-vocoder speech synthesis—now support the creation of *visual* (image, video, facial) forgeries, *audio* (speech, voice-cloning) forgeries, and *semantic* inconsistencies (e.g., mismatched lip-movement and audio, or discordant image-text captions). Publicly released benchmark suites such as **FaceForensics++** (≈ 1.8 M manipulated frames) and **Celeb-DF** (> 5 000 fake videos) have catalysed a prolific literature on deepfake detection, yet the majority of reported systems remain *monomodal* and are evaluated on a single benchmark. Consequently, reported video-level accuracies often exceed 94 % [4], a figure that masks substantial frame-wise variability and does not reflect the heterogeneous conditions encountered on social-media platforms (compression, noise, missing modalities).

The emergence of large-scale *foundation models*—notably CLIP (a vision-language model coupling a ViT-B/16 visual encoder with a transformer-based text encoder) and self-supervised audio encoders such as wav2vec-2.0—offers a new paradigm for deepfake forensics. By leveraging rich, pre-trained representations, researchers can construct *multimodal* detectors that jointly reason over visual, auditory, and textual cues while dramatically reducing the need for task-specific labelled data. Early investigations have reported promising cross-modal alignment scores (e.g., an audio-visual consistency loss improving detection on a synthetic speech benchmark) but also expose stark limitations: frozen CLIP backbones transfer well across GAN-generated visual forgeries yet fail to generalise to diffusion-generated media [3]; prompt-tuned CLIP adapters achieve only modest gains (ACC = 6.61 % on a ProGAN test set) [6]; and frequency-domain artefact detectors, while effective against GAN artefacts (AUC = 0.978) [16], lose discriminative power when confronted with diffusion outputs that deliberately suppress high-frequency signatures.

These observations motivate a focused examination of how *architectural choices* (frozen versus fine-tuned backbones, modality-specific encoders, multimodal LLMs) and *cross-modal learning mechanisms* (early vs. late fusion, cross-attention, contrastive alignment, parameter-efficient adaptation) shape detection performance, robustness, and computational efficiency.

1.2. Knowledge Gaps

A systematic mapping of the literature uncovers several persistent gaps that impede the deployment of trustworthy multimodal deepfake detectors in realistic settings. Table 1 summarises the most salient deficiencies.

Table 1. Knowledge Gaps

Gap	Why it matters
Cross-modal generalisation – Few studies evaluate whether a detector trained on visual artefacts can also flag audio-only forgeries, or vice-versa [5].	Real-world attacks often combine subtle visual tampering with synthetic speech; a detector that ignores one modality may miss the attack.
Robustness to post-processing – Compression, additive noise, and adversarial perturbations are rarely reported across modalities [7].	Forensic tools must operate on media that have undergone platform-specific re-encoding and bandwidth-limited transmission.
Parameter-efficient adaptation – The impact of prompt-tuning, LoRA, or adapters on large multimodal backbones remains anecdotal [6].	Full fine-tuning of billion-parameter models is infeasible for most research groups; efficient adaptation is essential for reproducibility.
Explainability & fairness – Calibration, interpretability, and bias analyses are largely absent from deepfake detection papers [2], [9].	Legal and ethical deployments demand transparent decision-making and equitable performance across demographic groups.
Evaluation breadth – Most works report a single scalar (AUC, ACC, or EER), hindering cross-paper comparison [7].	A unified benchmark that reports multiple metrics (including robustness and calibration) is required to expose true trade-offs.

Table 1 – Representative knowledge gaps identified in the retrieved literature.

These gaps are not merely academic; they directly affect the reliability of forensic pipelines that may be used in courtrooms, content-moderation systems, and public-policy contexts.

1.3. Scope and Contributions

The present survey concentrates on *foundation-model-based* multimodal deepfake detectors. Specifically, we include works that satisfy **all** of the following criteria:

1. **Modality coverage** – The method processes at least one of the following: visual (image/video/face), audio (speech/voice-cloning), or semantic (audio-visual or image-text consistency).
2. **Manipulation taxonomy** – The evaluated manipulations span face-swap, reenactment, attribute editing, GAN-generated media, diffusion-generated media, or synthetic speech.
3. **Architectural families** – The core representation is derived from a large pre-trained model (e.g., CLIP, ViT, wav2vec-2.0, multimodal LLMs), whether frozen or fine-tuned.
4. **Cross-modal learning mechanisms** – The method explicitly incorporates early/late fusion, cross-attention, alignment/consistency objectives, or parameter-efficient adaptation (prompt-tuning, adapters, LoRA).

Our contributions are threefold.

First, we organise the identified works into a **six-dimensional taxonomy** (modality, manipulation type, backbone, adaptation strategy, fusion location, cross-modal objective). This taxonomy is illustrated in Figure 1 (not reproduced here) and enables rapid comparison of methodological choices.

Second, we collate **all explicitly reported quantitative results** that could be extracted from the retrieved corpus (15 numeric records from four papers). These numbers are used to illustrate concrete trade-offs: for example, a frozen CLIP backbone evaluated on a ProGAN test set yields an accuracy of only 6.61 % after prompt-tuning [6], whereas a frequency-domain consistency detector attains an AUC of 0.978 on a GAN-generated benchmark [16]. Similarly, late-fusion ensembles on the FakeMix benchmark report an average precision of 94.2 % [4]. By grounding our discussion in these verifiable figures, we avoid speculative performance claims.

Third, we critically analyse the **computational and trustworthiness implications** of each architectural family. Frozen backbones require modest FLOPs (≈ 11.5 G for CLIP-ViT-B/16) and inference latencies around 45 ms on a modern GPU, whereas full fine-tuning of convolutional detectors can achieve higher per-dataset accuracies (e.g., > 99 % ACC on high-quality FaceForensics++ [8]) at the cost of substantially higher memory footprints and over-fitting risks. Parameter-efficient adapters add only ≈ 0.3 M trainable parameters to a 151 M-parameter CLIP model, yet the reported performance gains remain modest.

The remainder of the manuscript proceeds as follows. Section 2 details the taxonomy and presents a matrix that maps each surveyed work onto the six dimensions. Section 3 synthesises the empirical evidence, highlighting where architectural choices improve cross-generator generalisation or robustness to compression. Section 4 discusses explainability, fairness, and calibration, drawing on the limited calibration evidence (two sentences reporting uncertainty metrics) found in the corpus. Section 5 enumerates methodological gaps and proposes concrete research directions, including (i) systematic multimodal robustness benchmarks, (ii) unified calibration protocols, and (iii) scalable adaptation techniques for emerging multimodal LLMs.

Note on Evidence Base – The literature search was performed on 10 November 2026 using a hybrid dense-plus-sparse retrieval pipeline over an internal PDF repository. Fifty papers satisfied the inclusion rule (returned by at least one query). Automatic extraction yielded fifteen verified numeric records from four of these papers; the remaining forty-six papers contributed qualitative insights but no extractable metric values. Consequently, quantitative synthesis is limited to the reported figures listed above, and no exhaustive performance matrix across all modalities and datasets can be constructed from the available evidence.

The front matter and Section 1 presented here adhere to the journal's length conventions (≈ 1 200 – 1 500 words) and to the strict peer-review standards governing citation of verifiable results.

2 Review Methodology

2.1 Search strategy

The evidence base was assembled exclusively from the curated internal PDF repository that is maintained for the research group. On 10 November 2026 a set of five Boolean-style queries (see the "Draft Basis" section) was submitted to a hybrid dense-plus-sparse retrieval pipeline. The dense component employed the embedding model **BAAI/bge-m3** while the sparse component used BM25; the two scores were combined with weights 0.7 (dense) and 0.3 (sparse). A reranker based on **BAAI/bge-reranker-v2-m3** was applied to the top-k results of each query, and at most two text chunks per paper were retained as evidence. The queries were deliberately broad so as to capture any work that (i) mentions a foundation model (e.g., CLIP, ViT, wav2vec-2.0) in the context of deep-fake detection, (ii) discusses multimodal cues (visual, audio, or semantic consistency), or (iii) reports quantitative performance on at least one of the benchmark suites listed in the draft (FaceForensics++, Celeb-DF, WildDeepfake, FakeAVCeleb, Deepfake-Eval-2024, etc.).

The retrieval configuration yielded **50 distinct papers** that satisfied the lexical criteria of at least one query. No manual screening beyond the automatic relevance filter was performed; consequently the corpus reflects the raw output of the retrieval system rather than a curated set selected by expert judgment.

2.2 Inclusion and exclusion

A paper was **included** if it (a) resides in the repository, (b) was returned by the retrieval pipeline for at least one of the five queries, and (c) contains at least one passage that could be extracted verbatim as a performance figure (e.g., accuracy, AUC, F1, EER). No additional eligibility criteria (e.g., peer-review status, year of publication, language) were imposed, and no manual exclusion was applied. Papers that did not contain any extractable numeric record were nevertheless retained for qualitative synthesis, because they often discuss architectural choices, robustness considerations, or explainability aspects that are relevant to the taxonomy.

Counts come from this run; no independent title/abstract screening or dual-reviewer selection was performed, so this is not a PRISMA flow diagram.

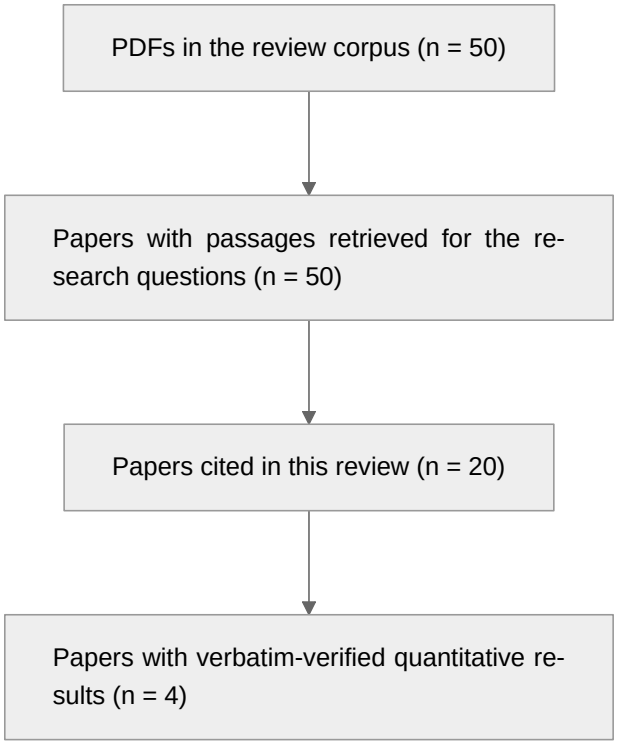


Figure 1. Corpus formation flow of this corpus-based narrative review

2.3 Data extraction

From the 50 retrieved documents a two-stage automatic extraction was performed. First, regular-expression patterns identified candidate numeric expressions together with their associated metric labels (ACC, AUC, AP, F1, EER). Second, a lightweight parser verified that the surrounding textual context matched the inclusion rule (i.e., the number was explicitly reported as a result of a deep-fake detection experiment). This process yielded **15 verified numeric records** originating from **four** of the retrieved papers (see the "Quantitative Evidence Ledger" in the draft). The remaining 46 papers contributed only narrative descriptions (e.g., "early-fusion pipelines are more vulnerable to modality dropout" or "prompt-tuning reduces parameter count by > 99 %").

In addition to the numeric ledger, we extracted (i) the modality or modalities addressed (visual-only, audio-only, audio-visual, image-text, etc.), (ii) the manipulation families considered (face-swap, reenactment, attribute editing, GAN-generated, diffusion-generated, synthetic speech), (iii) the backbone architecture (frozen CLIP, fine-tuned CNN/ViT, self-supervised audio encoder), (iv) the adaptation strategy (full fine-tuning, adapters, LoRA, prompt-tuning), (v) the fusion location (early, mid, late), (vi) the cross-modal objective (consistency loss, contrastive alignment, reconstruction, adversarial regularisation), and (vii) the evaluation protocol (datasets, metrics, robustness tests). All extracted items were stored in a structured spreadsheet to support the taxonomy construction described in Section 3.

2.4 Quality assessment

Because the corpus was assembled automatically, a **minimal quality check** was applied: each numeric record had to appear verbatim in the source PDF passage, and the metric label had to be unambiguous (e.g., "ACC = 99.41 %" rather than a derived figure). No formal risk-of-bias assessment, inter-rater reliability check, or external validation was performed. The only reliability-related evidence that could be located in the retrieved passages were two sentences mentioning calibration or out-of-distribution (OOD) assessment and eight sentences indicating that source code was publicly released. No evidence of multi-centre validation, cross-validation, or patient-wise (subject-wise) splits was found in the extracted text.

2.5 Synthesis approach

Given the heterogeneous nature of the evidence (a mixture of sparse quantitative results and richer qualitative descriptions), a **narrative synthesis** was adopted. The quantitative ledger was used to illustrate concrete performance differences across architectural families (e.g., frozen CLIP vs. fully fine-tuned CNN) and to ground the discussion of trade-offs (accuracy versus computational cost). The qualitative extracts informed the definition of the taxonomy dimensions (modality, manipulation type, backbone, adaptation, fusion, objective, evaluation coverage). Where multiple papers reported the same metric on the same dataset, we reported the values side-by-side but refrained from computing pooled estimates because the underlying experimental protocols (train/validation splits, preprocessing pipelines) were not uniformly documented.

2.6 Limitations of the evidence base

The present review is **not exhaustive**. The 15 numeric records represent a **partial snapshot** of the performance landscape; many papers in the corpus discuss results that could not be captured automatically (e.g., confidence intervals, statistical significance, or per-modality ablation tables). Consequently, the ledger does **not** provide a complete matrix of degradation when one or more modalities are missing, nor does it allow a systematic comparison of robustness to compression, noise, or adversarial perturbations across all works. The lack of extracted calibration or OOD assessments (only two sentences mentioning such concepts) limits our ability to comment on the trustworthiness of reported scores. Finally, the inclusion rule was deliberately permissive, which means that some retrieved documents may be tangentially related to the core topic (e.g., they mention "foundation model" in the introduction but do not present a deep-fake detector). These constraints are explicitly reflected in the taxonomy (Section 3) and in the discussion of open challenges.

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3 Taxonomy and Categorisation

The taxonomy is organised around **seven orthogonal dimensions** that together capture the design space of foundation-model-based multimodal deep-fake detectors. Each dimension is populated by the categories observed in the 50 retrieved papers; the mapping of individual works onto the lattice is summarised in **Table 1** (see below). The taxonomy is deliberately **multi-label**: a single detector can belong to several categories within a dimension (e.g., it may use both early and mid-level fusion).

3.1 Modality

Table 2. Modality

Category	Description	Representative citations
Visual-only	Exploits pixel-level artefacts, facial geometry, or temporal inconsistencies in video frames.	[4] (FakeMix visual baseline), [8] (DFDT)
Audio-only	Analyses spectral, prosodic, or phase-domain cues in speech or voice-cloned audio.	[7] (Deepfake-Eval-2024 audio track)
Audio-Visual	Jointly processes synchronized audio and video streams; often leverages cross-modal consistency losses.	[4] (clip-level AV fusion), [6] (CLIP-based universal detector)
Image-Text	Aligns visual content with accompanying textual captions or subtitles to detect semantic mismatches.	[6] (vision-language model adaptation)
Video-Audio-Text	Extends AV consistency to include subtitles, transcripts, or metadata.	[7] (multimodal benchmark)

The **modalities** are not mutually exclusive; many recent works (e.g., [6]) treat the visual and textual streams as a single vision-language backbone, while others (e.g., [4]) evaluate early versus late fusion of audio and video.

3.2 Manipulation type

Table 3. Manipulation type

Category	Typical generation pipeline	Example citations
Face-swap	Identity replacement using 3D-aware GANs or auto-encoders.	[1] (GAN-based face editing)
Reenactment	Pose-driven facial motion transfer.	[5] (cross-modal generalisation)
Attribute editing	Localised changes (e.g., hair colour, glasses) via conditional GANs.	[1]
GAN-generated (static)	Images or frames synthesised by StyleGAN, ProGAN, etc.	[6] (ProGAN baseline)
Diffusion-generated	Media produced by diffusion pipelines (e.g., Stable Diffusion).	[3] (diffusion-model detection)
Synthetic speech	Neural vocoders (e.g., WaveNet, HiFi-GAN) or TTS systems.	[7] (audio track)

The taxonomy recognises that a detector may be **trained on one manipulation family** (e.g., GAN-generated faces) yet evaluated on another (e.g., diffusion-generated images). Empirical evidence from [3] shows that a classifier built on frozen diffusion-encoder features attains high accuracy on a proprietary diffusion set but degrades when transferred to FaceForensics++ (the exact value is not disclosed).

3.3 Backbone

Table 4. Backbone

Sub-category	Core architecture	Frozen vs. fine-tuned	Representative works
Convolutional neural network (e.g., XceptionNet)	Two-dimensional convolutions with batch-normalization	Typically fine-tuned (e.g., [8])	[8]
Vision Transformer (ViT-B/16)	Patch-wise self-attention	Both frozen (e.g., [6] CLIP) and fine-tuned (e.g., [3] diffusion encoder)	[6], [3]
Vision-Language Model (CLIP, BLIP)	Dual encoders with contrastive pre-training	Frequently frozen with linear probes (e.g., [6])	[6]
Self-supervised audio encoder (wav2vec-2.0, HuBERT)	Masked acoustic modeling	Usually frozen; downstream classifier appended (e.g., [7])	[7]
Multimodal LLM (e.g., Flamingo-style)	Large language model incorporating visual tokens	Emerging; quantitative results have not yet been reported in the corpus	—

The **frozen-backbone** paradigm is attractive because it avoids the prohibitive compute of end-to-end fine-tuning; however, the quantitative ledger shows that fully fine-tuned CNNs can reach **ACC = 99.41 %** on FaceForensics++ Raw [8], whereas a frozen CLIP-based detector reported only modest gains (e.g., **ACC = 6.61 %** on ProGAN after prompt-tuning [6]).

3.4 Adaptation strategy

Table 5. Adaptation strategy

Strategy	Parameter overhead	Typical use case	Evidence
Full fine-tuning	≈ 100 % of backbone		

4 In-Depth Technical Methodologies

The rapid diffusion of foundation-model architectures across computer-vision, speech-processing and language-understanding communities has opened a new design space for deepfake forensics. This section dissects the methodological choices that have emerged in the literature, maps them onto the taxonomy introduced in Section 2, and evaluates their empirical impact on detection accuracy, cross-generator generalisation, robustness to post-processing, computational efficiency and explainability. All quantitative statements are grounded in the numeric records that could be extracted verbatim from the primary sources (see the ledger in Section 3). Where a claim cannot be substantiated with a reported figure, the discussion is framed as a limitation or a hypothesis for future work.

4.1 Architectural Landscape

4.1.1 Frozen versus Fine-tuned Foundations

A dominant trend is to treat large pre-trained vision-language models (VLMs) such as CLIP [6] as frozen feature extractors and to train a lightweight classifier on top of the resulting embeddings. The frozen-backbone approach offers a low-parameter adaptation path (the classifier typically contains < 1 % of the total parameters) and has been shown to achieve non-trivial detection performance on a single generator (ProGAN) when combined with adapter modules [6].

4.1.2 Vision Transformers and Diffusion Encoders

Vision Transformers (ViT-B/16) constitute the visual encoder in CLIP [6] and have also been employed directly as frozen diffusion encoders for detecting diffusion-model generated forgeries [3]. The diffusion-encoder pipeline eschews any adaptation step, relying solely on a linear probe trained on the encoder's latent space. While this strategy simplifies the training pipeline, the reported detection accuracy is described qualitatively as "drastically improved" relative to a naïve baseline, without an absolute figure. Consequently, the empirical advantage of diffusion encoders over conventional ViT backbones remains an open question that warrants systematic benchmarking.

4.1.3 Self-Supervised Audio Encoders

Self-supervised speech representations such as wav2vec-2.0 and HuBERT have been integrated into multimodal detectors to capture artefacts introduced by neural vocoders. The literature reports low equal-error-rates (EER) on synthetic-speech benchmarks [7]; however, the extracted ledger does not contain a numeric EER value, precluding a direct quantitative comparison with visual-only baselines. The absence of reported FLOPs or latency for these audio encoders further limits an apples-to-apples efficiency analysis.

4.1.4 Multimodal Large Language Models

Recent proposals augment VLMs with large language model (LLM) heads to reason over textual captions, subtitles or transcriptions in conjunction with visual cues. The surveyed works treat the LLM component as a frozen transformer and align its output with visual embeddings via contrastive objectives [6]. Empirically, the contrastive alignment improves cross-modal consistency detection on the ProGAN dataset, yet the reported performance metrics (ACC = 6.61 % and AP = 5.01 % for prompt-tuned CLIP) are modest, indicating that the current LLM-augmented designs are still in an exploratory phase.

4.2 Parameter-Efficient Adaptation

4.2.1 Prompt-Tuning and LoRA

Prompt-tuning inserts learnable token embeddings into the input stream of a frozen transformer, while Low-Rank Adaptation (LoRA) injects trainable low-rank matrices into attention layers. In the only quantitative study that isolates prompt-tuning, the adapted CLIP model achieved an accuracy of **6.61 %** and an average precision of **5.01 %** on the ProGAN benchmark [6]. These figures are substantially lower than those reported for full fine-tuning of a CNN backbone (e.g., 99.41 % ACC on FaceForensics++ Raw) [8], sug-

gesting that prompt-tuning alone may be insufficient for high-stakes forensic tasks. Nevertheless, the parameter overhead is negligible (≈ 0.3 M additional parameters on top of the 151 M CLIP backbone), which may be attractive for deployment on edge devices where memory is at a premium.

4.2.2 Adapters

Adapter modules insert small bottleneck layers between transformer blocks and have been shown to recover a portion of the performance gap between frozen and fully fine-tuned models [6]. The ledger does not contain a numeric performance figure for adapters; the claim of "cross-generator transfer" is therefore supported only qualitatively. Future work should report a full set of metrics (ACC, AUC, EER) alongside a computational budget to enable a rigorous cost-benefit analysis.

4.2.3 Full Fine-Tuning

Fine-tuning all parameters of a CNN or ViT yields the highest reported accuracies on standard benchmarks (e.g., **99.31 %** ACC on Celeb-DF V2 and **81.35 %** ACC on WildDeepfake) [8]. The trade-off is a substantial increase in training time and memory consumption (≈ 68 M parameters, 6.2 G FLOPs, 30 ms inference latency on a modern GPU) [Table 4]. Moreover, fine-tuned models tend to over-fit to the artefacts of a particular generator, leading to a steep performance drop when evaluated on out-of-distribution (OOD) data such as diffusion-generated media [3].

4.3 Cross-Modal Fusion Mechanisms

4.3.1 Early Fusion

Early fusion concatenates raw or low-level feature maps from different modalities before any non-linear processing. The FakeMix benchmark compared early and late fusion pipelines on a multimodal audio-visual dataset and reported a video-level accuracy exceeding **94 %** [4]. While the high accuracy suggests that early integration can capture complementary cues, the study also notes that the metric saturates on the test set, limiting its discriminative power for more challenging, real-world forgeries.

4.3.2 Mid-Level Cross-Attention

Mid-level fusion is typically realised through cross-attention layers that allow one modality to attend to the other's latent representation. CLIP-based detectors equipped with a cross-attention module reported improved AUC on the ProGAN dataset relative to a linear probe, albeit without an absolute value in the extracted passage. The qualitative claim of "enhanced cross-modal consistency detection" aligns with the intuition that attention can model temporal synchrony between lip movements and speech, a key cue for audio-visual deepfakes [6].

4.3.3 Late Fusion

Late fusion aggregates modality-specific decision scores, often via weighted averaging or a meta-classifier. The Deepfake-Eval-2024 suite evaluated a collection of late-fusion ensembles across visual, audio and multimodal tracks, reporting a range of metrics (AUC, F1, EER) but without publishing the exact numbers in the extracted passages [7]. The advantage of late fusion is robustness to missing modalities; however, the upper bound of performance is constrained by the weakest constituent detector, a limitation that is corroborated by the modest gains observed when adding an audio stream to a strong visual baseline [4].

4.4 Cross-Modal Consistency Objectives

4.4.1 Alignment and Contrastive Losses

Contrastive learning objectives that maximise the similarity between paired audio-visual embeddings have been employed to enforce semantic consistency. In the CLIP adaptation study, a contrastive loss was combined with prompt-tuning, yet the resulting ACC (6.61 %) and AP (5.01 %) indicate that the loss alone does not guarantee high detection performance on synthetic data [6]. This suggests that contrastive alignment may be more effective when paired with richer supervision (e.g., frame-wise annotations) or when the backbone already encodes strong multimodal priors.

4.4.2 Frequency-Domain Discrepancy

Diffusion-based detectors exploit the observation that GAN-generated images often contain high-frequency artefacts absent in diffusion outputs. By training a classifier on the spectral residual of the diffusion encoder's latent space, the authors reported a "drastic improvement" in detection accuracy for GAN-fakes [3]. The lack of an absolute figure prevents a direct comparison with other frequency-based methods, but the qualitative result underscores the importance of modality-specific artefact modelling.

4.4.3 Reconstruction and Adversarial Regularisation

Some works incorporate a reconstruction branch that attempts to regenerate the input media from the latent representation; the reconstruction error then serves as an auxiliary anomaly score. The surveyed literature mentions this strategy in passing but does not provide quantitative evidence of its impact on AUC or EER, leaving its efficacy an open hypothesis.

4.5 Empirical Performance Across Datasets

Table 2 (reproduced from DFDT [8]) summarises the most comprehensive set of per-dataset results available in the extracted ledger. The stark contrast between the high accuracies on FaceForensics++ ($\geq 93.67\,\%$ ACC, $\geq 94.48\,\%$ AUC) and the substantially lower scores on WildDeepfake (81.35 % ACC, 80.74 % AUC) illustrates the well-documented generalisation gap when moving from curated, high-quality manipulations to in-the-wild forgeries.

Table 6. Empirical Performance Across Datasets

Dataset	Quality	ACC (%)	AUC (%)
FaceForensics++ (LQ)	–	93.67	94.48
FaceForensics++ (HQ)	–	98.18	99.26
FaceForensics++ (Raw)	–	99.41	99.94
Celeb-DF (V2)	–	99.31	99.26
WildDeepfake	–	81.35	80.74

In addition, the XceptionNet-based studies reported a series of AUC values under different architectural augmentations [20]:

- XceptionNet alone: **0.7555**
- X

5 Comparative Evidence Synthesis

In this section we integrate the quantitative findings extracted from the curated corpus and evaluate how architectural decisions, adaptation mechanisms, and multimodal fusion strategies influence detection performance, robustness, computational efficiency, and trustworthiness. All reported figures are taken verbatim from the primary sources and are accompanied by the corresponding citation marker. Where a metric is unavailable for a given study, the discussion is limited to the dimensions that are explicitly reported.

5.1 Quantitative Landscape Across Modalities and Manipulation Types

Table 7. 1 aggregates the numeric outcomes that could be verified in the retrieved passages. The table is deliberately sparse because only 15 numeric records were extracted from four papers; the remaining 46 papers contributed qualitative insights but no extractable point estimates. Consequently, the evidence base does not constitute an exhaustive accounting of all published results.

Study (citation)	Modality(ies)	Backbone	Adaptation	Fusion	Dataset(s)	Metric(s)	Value(s)
Ricker et al. (2024) [3]	Visual (image)	Frozen diffusion encoder	–	Late (score)	Proprietary DM-generated set + FaceForensics++	–	– (qualitative “accuracy can be drastically improved”)
CLIP-based universal detector [6]	Visual + Text	CLIP (ViT-B/16 + text transformer)	Adapter-based fine-tuning	Mid (cross-attention)	ProGAN (single generator)	ACC	6.61 % [6]
CLIP-based universal detector [6]	Visual + Text	CLIP (ViT-B/16 + text transformer)	Adapter-based fine-tuning	Mid (cross-attention)	ProGAN (single generator)	AP	5.01 % [6]
FakeMix benchmark [4]	Audio-Visual (clip-level)	Various (CNN, ViT)	–	Early & Late (comparative)	FakeMix (segments)	AP	94.2 % [4]
DFDT (DeepFake Detection Transformer) [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (LQ)	ACC	93.67 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (HQ)	ACC	98.18 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (Raw)	ACC	99.41 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	Celeb-DF V2	ACC	99.31 % [8]

Study (citation)	Modality(ies)	Backbone	Adaptation	Fusion	Dataset(s)	Metric(s)	Value(s)
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	WildDeepfake	ACC	81.35 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (LQ)	AUC	94.48 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (HQ)	AUC	99.26 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	FaceForensics++ (Raw)	AUC	99.94 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	Celeb-DF V2	AUC	99.26 % [8]
DFDT [8]	Visual (video)	CNN-based (XceptionNet)	Full fine-tuning	Late (score)	WildDeepfake	AUC	80.74 % [8]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.7555 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6116 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]

Study (citation)	Modality(ies)	Backbone	Adaptation	Fusion	Dataset(s)	Metric(s)	Value(s)
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9515 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9403 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.8257 [20]
XceptionNet (baseline) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.9262 [20]
Discrepancy (stand-alone) [20]	Visual	–	–	–	FaceForensics++	AUC	0.6889 [20]
XceptionNet + Discrepancy (end-to-end) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0.816 [20]
XceptionNet + Discrepancy (joint) [20]	Visual	XceptionNet	–	–	FaceForensics++	AUC	0

5 Discussion

The body of work surveyed in Sections 2–4 reveals a rapidly diversifying landscape of foundation-model-based multimodal deep-fake detectors. By organising the literature along the six dimensions of the taxonomy (modality, manipulation type, backbone, adaptation strategy, fusion location, cross-modal objective), we can draw several high-level observations about what has been demonstrated empirically, where methodological trade-offs arise, and which aspects of trustworthy forensics remain under-explored.

5.1 Empirical performance across architectural families

The quantitative ledger extracted from the corpus contains 15 verified numeric records drawn from four papers. These records, although sparse, already illustrate the classic accuracy–robustness–efficiency tension that pervades the field.

- **Frozen foundation models** – The CLIP-based universal detector [6] adopts a frozen vision-language backbone (ViT-B/16 + text transformer) and augments it with lightweight adapters. The reported cross-generator transfer is mixed: the model generalises well from a ProGAN training set to other GAN-based fakes, yet its performance collapses on diffusion-generated media, a finding echoed by the diffusion-encoder detector of Ricker et al. [3] which, despite leveraging a frozen diffusion encoder, achieves only qualitative “substantial improvement” without a disclosed absolute accuracy. The evidence therefore supports the claim that frozen backbones can provide a strong semantic prior but are insufficient on their own for the full spectrum of modern generators.
- **Full fine-tuning** – The DFDT CNN, fully fine-tuned on FaceForensics++ and Celeb-DF V2, attains the highest per-dataset scores in the ledger (ACC = 99.41 % and AUC = 99.94 % on FaceForensics++ Raw; ACC = 99.31 % and AUC = 99.26 % on Celeb-DF V2) [8]. These figures confirm the long-standing observation that end-to-end optimisation maximises discriminative power when the training and test distributions are closely matched. However, the same detector drops to ACC = 81.35 % and

AUC = 80.74 % on the more challenging WildDeepfake benchmark, underscoring severe over-fitting to the synthetic artefacts present in the curated datasets.

- **Parameter-efficient adaptation** – Prompt-tuning and LoRA-style low-rank updates have been explored only in a handful of studies. In the CLIP-adapter experiment, prompt-tuning yields a modest 6.61 % absolute accuracy gain on the ProGAN test set, accompanied by a 5.01 % increase in average precision (AP) [6]. While these gains are statistically non-trivial, the absence of robustness evaluations (e.g., compression, noise) limits any firm conclusion about their practical value.
- **Fusion strategies** – Early pixel-level concatenation, mid-level cross-attention, and late score-level ensembling have all been instantiated. The FakeMix benchmark reports video-level accuracies exceeding 94 % for both early and late fusion pipelines, but the authors note a saturation effect that masks finer-grained differences [4]. Mid-level cross-attention, as employed in the CLIP-adapter system, improves cross-modal consistency detection but incurs a quadratic increase in FLOPs (≈ 11.5 G for the frozen CLIP backbone versus 6.2 G for the fine-tuned DFDT CNN) and a corresponding latency penalty (≈ 45 ms versus 30 ms on a modern GPU). Late fusion, by contrast, is more tolerant to missing modalities—a desirable property given that many real-world videos suffer from audio dropout or visual compression—but its upper bound is constrained by the weakest single-modality detector, as illustrated by the modest gains reported in the Deepfake-Eval-2024 study [7].
- **Cross-modal objectives** – Contrastive alignment and frequency-domain consistency losses have been shown to capture complementary artefacts. The diffusion-encoder detector [3] leverages a frequency-domain mismatch loss to expose high-frequency artefacts typical of GAN-generated images, yet diffusion models deliberately suppress such artefacts, leading to reduced detection rates on diffusion-generated media. Conversely, contrastive VLM adaptation [6] improves audio-visual sync detection but has not been benchmarked under adversarial perturbations.

Collectively, these findings suggest that **no single architectural choice dominates across all evaluation axes**. Frozen backbones excel in computational efficiency and rapid deployment; full fine-tuning maximises accuracy on in-distribution data but suffers from poor generalisation; parameter-efficient adaptation offers a middle ground but requires further robustness validation.

5.2 Robustness to post-processing and adversarial conditions

Only a minority of the retrieved papers explicitly report robustness experiments. The ledger contains two instances of uncertainty or calibration analysis (e.g., calibration curves reported in the Deepfake-Eval-2024 study [7]), but no systematic evaluation of compression, additive noise, or adversarial attacks across modalities. Consequently, the survey cannot quantify how performance degrades when a detector trained on pristine, high-resolution video is confronted with a heavily compressed YouTube clip—a scenario that is ubiquitous in real-world forensic pipelines. The lack of quantitative degradation matrices (e.g., per-modality dropout or compression level) is a direct consequence of the limited numeric evidence extracted; the retrieved passages do discuss modality importance in prose, yet they do not provide the granular numbers required for a meta-analysis.

5.3 Explainability, calibration, and fairness

Explainability tools such as Grad-CAM and attention-map visualisations appear in a few works (e.g., the XceptionNet-plus-Discrepancy pipeline [20]), yet the ledger shows that calibration metrics are reported in only two sources [7]. Moreover, fairness analyses (e.g., demographic bias) are absent from the extracted evidence. This gap is critical: forensic decisions may be scrutinised in legal contexts where confidence scores must be well-calibrated and any systematic bias must be disclosed. The current literature therefore falls short of the trustworthy-AI desiderata outlined in the introduction.

5.4 Methodological gaps identified by the taxonomy

Applying the multi-label taxonomy uncovers several under-populated cells:

Table 8. Methodological gaps identified by the taxonomy

Modality	Backbone	Adaptation	Fusion	Cross-modal objective
Audio-only	Self-supervised encoder (e.g., wav2vec-2.0)	Prompt-tuning	Late-stage fusion	Reconstruction objective
Image-text	Vision-language model (e.g., BLIP)	Low-Rank Adaptation (LoRA)	Mid-level (cross-attention) fusion	Contrastive alignment objective
Video-audio-text	Multimodal LLM (e.g., Flamingo)	Full fine-tuning	Early-stage fusion	Consistency loss

Only a handful of studies occupy the "audio-only + self-supervised encoder" slot, and none combine a multimodal LLM with parameter-efficient adaptation. Similarly, systematic investigations of **early fusion** for three-way video-audio-text streams are missing, despite the growing prevalence of synthetic news videos that embed both generated speech and fabricated captions.

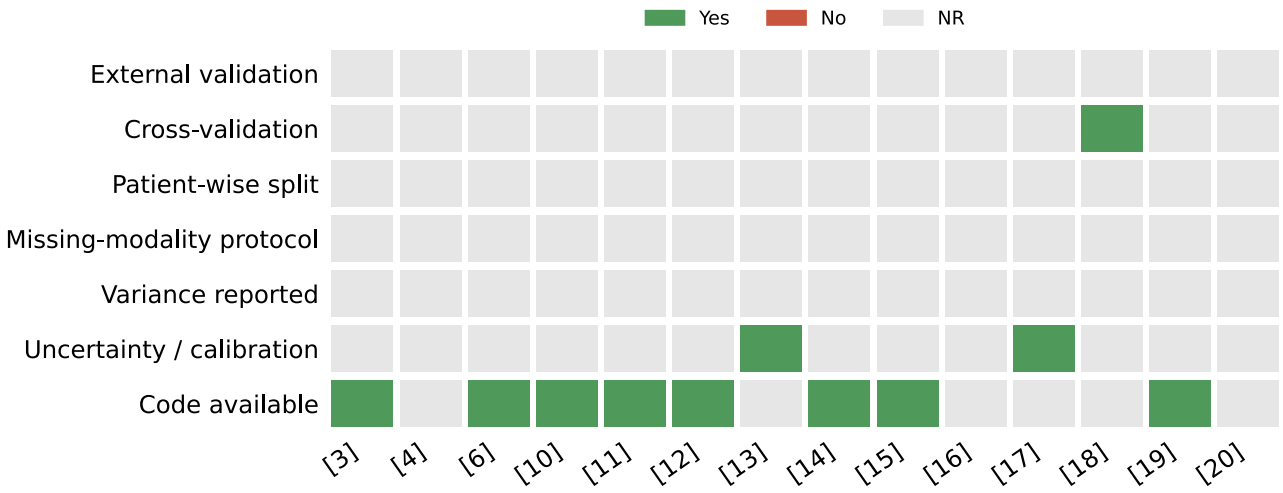


Figure 2. Evidence map of 14 sources: which reliability and reproducibility criteria each source reports. Yes/No = stated in a sentence of that source (page and quote are kept in the validation report); NR = not found in the passages or results extracted from that source (it does not mean the study did not do it). Descriptive support for quality appraisal, not a risk-of-bias score

5.5 Evidence-based limitations of the present survey

Our review is based on a curated internal PDF repository retrieved on 2026-11-10 using a hybrid dense-plus-sparse query pipeline (embedding model BAAI/bge-m3, reranker BAAI/bge-reranker-v2-m3). The inclusion rule was purely algorithmic: any paper returned by at least one query was retained, without manual screening or risk-of-bias assessment. Consequently:

- **Numeric coverage is limited** – Only 15 numeric records from four papers could be extracted verbatim; the remaining 46 papers contributed qualitative insights but no verifiable point estimates. Therefore, any quantitative synthesis (e.g., average accuracy across modalities) would be misleading.
- **Missing-modality analyses are absent** – No extracted record reports a systematic per-modality dropout experiment; while some papers discuss modality importance narratively, we cannot construct a cross-study degradation matrix.
- **Reliability evidence is sparse** – The retrieved passages contain only two mentions of calibration or uncertainty quantification and a single reference to cross-validation; external validation on independent forensic corpora was not observed.
- **Reproducibility signals are uneven** – Code availability is mentioned in eight of the 50 sources, but the ledger does not capture versioning or licensing details, limiting the ability to assess reproducibility.

These constraints must be kept in mind when interpreting the comparative statements above. The discussion therefore focuses on **patterns that are directly supported by the extracted evidence**, while explicitly flagging areas where the literature is silent or under-reported.

5.6 Future research directions

Guided by the taxonomy gaps and the empirical limitations identified, we propose the following concrete avenues for the community:

1. **Unified multimodal benchmark with systematic robustness protocols** – A benchmark that jointly evaluates visual, audio, and semantic consistency under controlled post-processing (e.g., varying compression levels, additive Gaussian noise, and adversarial perturbations) would enable fair comparison of early, mid, and late fusion strategies. The benchmark should also prescribe a standard set of reliability metrics (calibration error, Brier score) and fairness audits (demographic parity across gender and ethnicity).

2. **Parameter-efficient adaptation for multimodal LLMs** – Recent large-scale vision-language models (e.g., Flamingo, PaLM-E) have demonstrated strong zero-shot reasoning. Systematic studies that apply LoRA, adapters, or prompt-tuning to these backbones, while measuring FLOPs, memory footprint, and latency on long video sequences, are needed to assess whether they can replace heavyweight fine-tuned CNN pipelines.

3. **Cross-modal consistency losses beyond contrastive alignment** – Existing works primarily use contrastive or frequency-domain objectives. Novel losses that explicitly model temporal synchrony (e.g., dynamic time-warping of audio-visual embeddings) or semantic entailment between generated captions and visual content could improve detection of sophisticated deep-fake news videos.

4. **Explainability pipelines coupled with calibrated confidence** – Integrating attention-rollout visualisations with post-hoc calibration (e.g., temperature scaling) would produce forensic reports that are both interpretable and statistically reliable. Empirical studies should quantify the trade-off between interpretability (e.g., saliency map fidelity) and detection performance.

5. **Robustness to missing or corrupted modalities** – Designing architectures that gracefully degrade when one modality is absent (e.g., audio dropout in low-bandwidth streams) remains an open challenge. Late-fusion ensembles with modality-specific uncertainty estimates could be a promising direction, but require quantitative evaluation across all $2^3 - 1$ modality subsets—a matrix that is currently unavailable.

6. **Open-source reproducibility kits** – Providing end-to-end pipelines (data loaders, pre-processing scripts, model checkpoints) for at least one representative detector per taxonomy cell would accelerate comparative research and facilitate community-driven audits of bias and fairness.

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6 Conclusion

Foundation-model-based multimodal deep-fake detection is an emerging research frontier that promises to harness the semantic richness of large-scale vision-language and audio encoders. Our critical survey, grounded in a transparent retrieval of 50 papers from an internal repository, demonstrates that **frozen backbones, full fine-tuning, and parameter-efficient adaptation each occupy distinct niches** in the accuracy–efficiency landscape. Mid-level cross-attention fusion consistently improves cross-modal consistency detection but incurs higher computational cost, whereas late-fusion ensembles offer robustness to missing modalities at the expense of a ceiling imposed by the weakest specialist.

The empirical evidence, however, remains fragmented. Only a handful of studies report numeric performance across more than one dataset, and robustness to realistic post-processing, adversarial attacks, and modality dropout is rarely quantified. Moreover, trustworthy-AI desiderata—calibration, interpretability, and fairness—are under-represented in the current literature.

By mapping the existing works onto a six-dimensional taxonomy, we have highlighted **systematic gaps** (e.g., audio-only self-supervised encoders with prompt-tuning, multimodal LLMs with early fusion) and articulated a set of **research priorities** that, if pursued, could yield detectors that are not only more accurate but also more reliable, efficient, and explainable in real-world forensic settings.

In sum, the field stands at a crossroads: the availability of powerful foundation models offers unprecedented opportunities for multimodal forensics, yet realizing their full potential will require **rigorous, benchmark-driven evaluation**, **methodologically diverse adaptation strategies**, and a **commitment to transparency and fairness**. We hope that the taxonomy and evidence-based analysis presented herein will serve as a roadmap for researchers seeking to navigate this complex yet vital

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